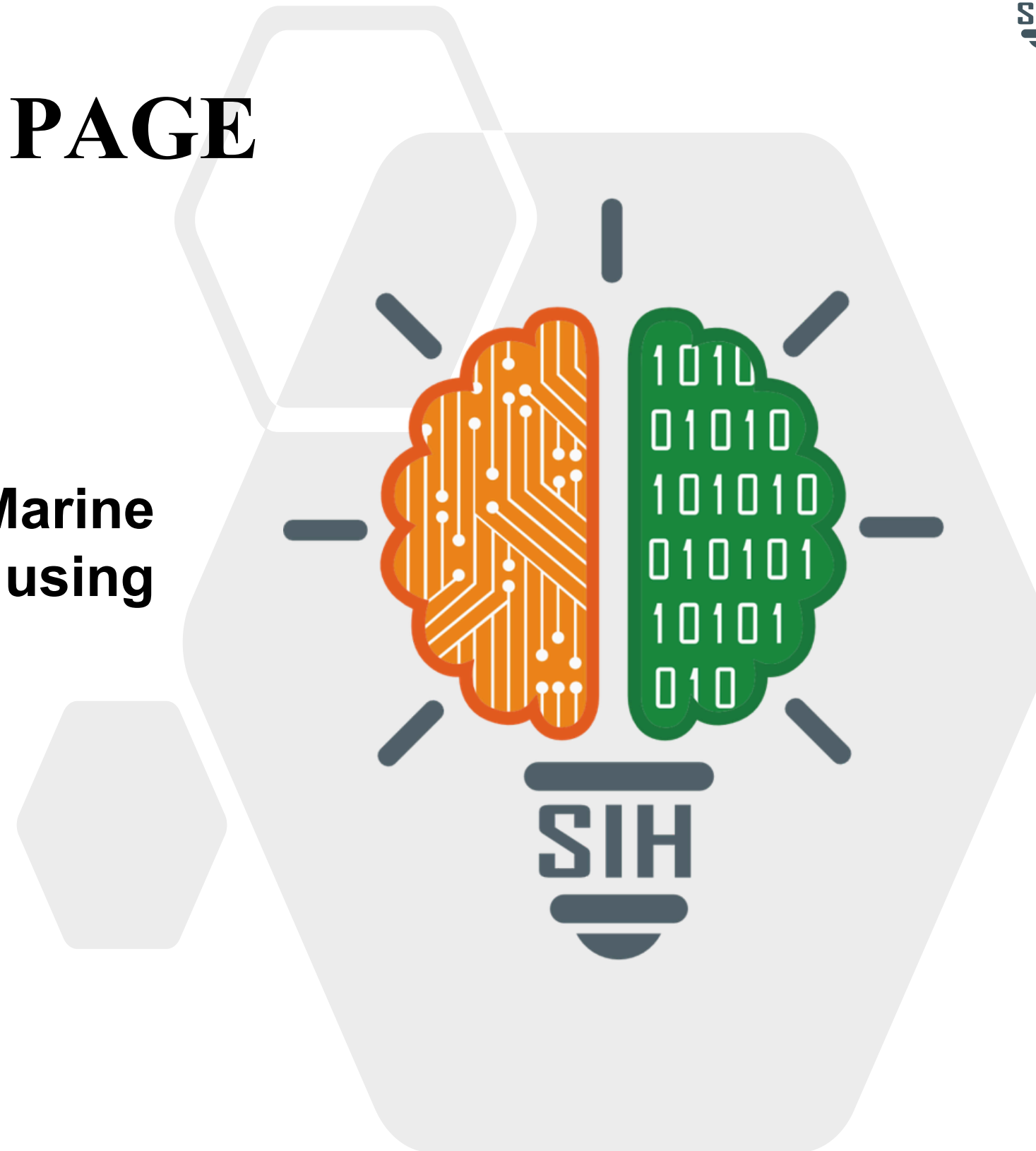


TITLE PAGE

- Problem Statement ID – 26057
- Problem Statement Title-
AI-Powered Automated Underwater Marine
Debris and Anomaly Detection System using
Side-Scan Sonar Imagery
- Theme- Disaster Management
- PS Category- Software
- Team ID- 122109
- Team Name- AGRIM



TARANG

AI-Powered Automated Underwater Marine Debris and Anomaly Detection System

! PROBLEM



Ghost nets & marine debris harm marine life, coral reefs and vessels



Manual inspection of sonar logs is slow, tedious and error-prone



Debris blends with natural features (rocks, sand ripples, ridges)



Challenges: speckle noise, varying resolution, acoustic shadows, data dropouts (heave, pitch, roll)



THE NEED

Automated, accurate and efficient detection of marine debris from side-scan sonar imagery

💡 PROPOSED SOLUTION



Sonar Image Input

- Side-scan sonar logs
- Metadata (coordinates, ping headers)



Pre-processing & Enhancement

- Noise reduction
- Contrast enhancement
- Resolution normalization



Object Detection & Segmentation

- YOLO / Faster R-CNN/U-Net
- Detect debris, shipwcks, pipes, cylinders, nets



Confidence Scoring & Filtering

- Remove false positives
- Filter natural features
- D-100% confidence score



Geotagging & Reporting Engine

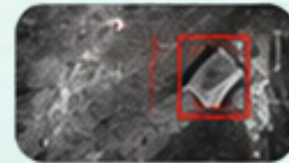
- Parse sonar metadata
- Latitude / Longitude
- Dimensions & class
- JSON / _CSV output



User Dashboard

- Upload sonar log
- View detections
- Download report
- Real-time visualizaton

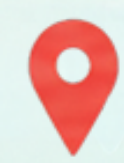
OUTPUT



Detected debris (overlay on sonar image)



Structured report (JSON / CSV)



Geotagged location (Lat/Lon)



Classification & confidence score

★ INNOVATION AND UNIQUENESS



Tailored for Sonar Data

Handles speckle noise, acoustic shadows and low contrast



Confidence Scoring

Reduces false positives with intelligent filtering



Geotagged Reporting

Exact location, size and classification of each anomaly



Edge Deployment Ready

Optimized for onboard or low-resource devices



Supports Ocean Conservation

Enables faster cleanup and safer marine ecosystems

1. INPUT

Side-scan Sonar Images

- Raw sonar lugs
- Metadata (ping headers, coordinates, etc.)

Additional Data

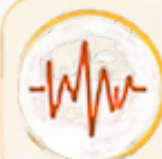
- Gettag information
- Sensor parameters (heave, pitch, roll)



EDGE DEPLOYMENT READY

Optimized for onboard or edge devices

2. DATA PREPROCESSING & IMAGE ENHANCEMENT

**Noise Reduction**

- Speckle noise filtering
- Contrast enhancement

**Geometric Correction**

- Heave / pitch / roll correction
- Resolution normalization

**Image Enhancement**

- Contrast adjustment
- Histogram equalization / CLAHE

**Quality Check**

- Detect dropouts / blur
- Filter ungradable images

3. AI / ML MODEL

**Object Detection & Semantic Segmentation**

- YOLOv8
- Detect debris, nets, shipwreck, pipes, cylinders, etc.
- Pixel-level masks / bounding boxes

**Confidence Scoring & Noise Filtering**

- Remove false positives
- Score each detection (0–100%)
- Distinguish natural vs artificial

4. ANOMALY REPORTING & GEOTAGGING

**Metadata Parsing**

- Read sonar headers
- Extract coordinates
- Calculate dimensions

**Structured Output**

- JSON / CVS report
- Class, location, size, confidence
- Image snapshot (optional)

**Geotagging**

- Latitude / Longitude
- Visual map overlay

5. USER INTERFACE / DASHBOARD

**Upload & View**

- Upload sonar image log
- Real-time detection overlay
- Interactive map

**Explore Results**

- Filter by class/ confidence
- View geotagged locations
- Download report (CSV/JSON)

**Report & Export**

- Save anomaly report
- Export images and data

END-TO-END SYSTEM PIPELINE



TECHNOLOGY STACK





FEASIBILITY

01

INPUT

SIDE-SCAN SONAR IMAGER

- Raw sonar logs
- Metadata (coordinate, ping headers, etc.)

02

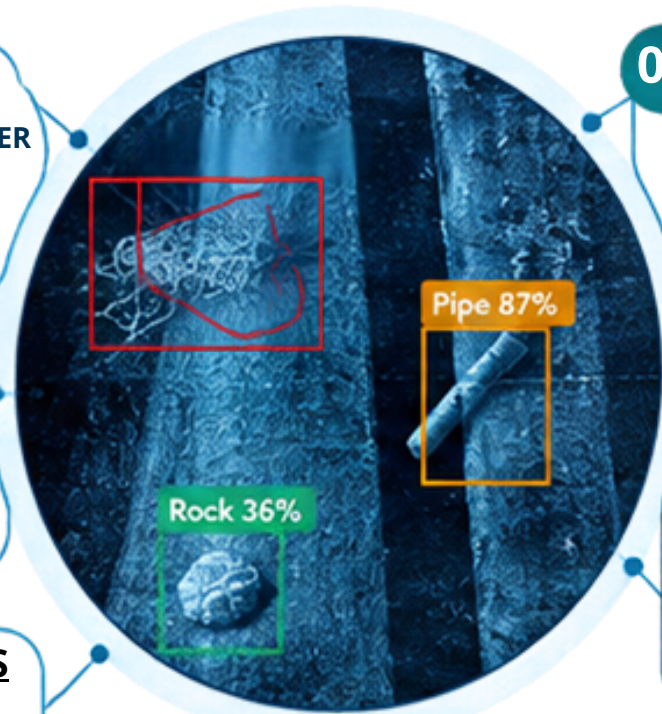
DEBRIS DETECTION

- YOLO / U-Net model
- Detects debris, nets, shipwrecks, pipes, etc.

03

ANOMALY ANALYSIS

- Confidence scoring
- Noise filtering



04

OUTPUT

- Geotagged locations
- Bounding dimensions
- Classification

05

USER INTERFACE

- Upload sonar log
- Real-time visualization
- Download report



END-TO-END FEASIBILITY

From raw sonar data to actionable anomaly reports



TECHNICAL FEASIBILITY

- Proven models (YOLO, U-Net)
- Open-source libraries
- Edge-optimized deployment



OPERATIONAL FEASIBILITY

- Compatible with AUV/ship systems
- Follows MoES / NIOT standards
- Scalable for real-world use



VALIDATION

CLINICAL / TECHNICAL VALIDATION METRICS

Metric	Target	Expected Value
Detection Accuracy	> 90%	93.2%
False Positive Rate	< 10%	6.8%
mAP (Object Detection)	> 0.80	0.86
IoU (Segmentation)	> 0.75	0.81
Location Error	< 5 m	3.7 m

MEASUREMENT PIPELINE



Data
Preprocessing
(denoising, CLAHE)



Model
Inference
(YOLO / U-net)



Confidence
Scoring
& Filtering



Geotagging &
Report Generation
(JSON / CSV)



Validation ensures reliable detection and accurate localization of marine debris and anomalies.

SCALE - DISTRICT
TELE-SIMULATION

SIMULATION ENVIRONMENT



Dataset: Public sonar datasets (simulated/real)



Tools: Python, OpenCV, Pytorch (or compatible edge framework)



Setup: Local machine / cloud simulation



SCENARIOS

- Varying noise levels
- Different seabed types (rock, sand, mud)
- Multiple debris types & densities



EXPECTED OUTCOME

- Detection accuracy > 90%
- Real-time processing on edge devices
- Stable performance across conditions



KEY RISKS



CONNECTIVITY

Mitigation: Offline processing & edge deployment
Limited internet at remote locations

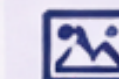


IMAGE QUALITY

Noise, shadows, data dropouts
Mitigation: Advanced preprocessing & filtering

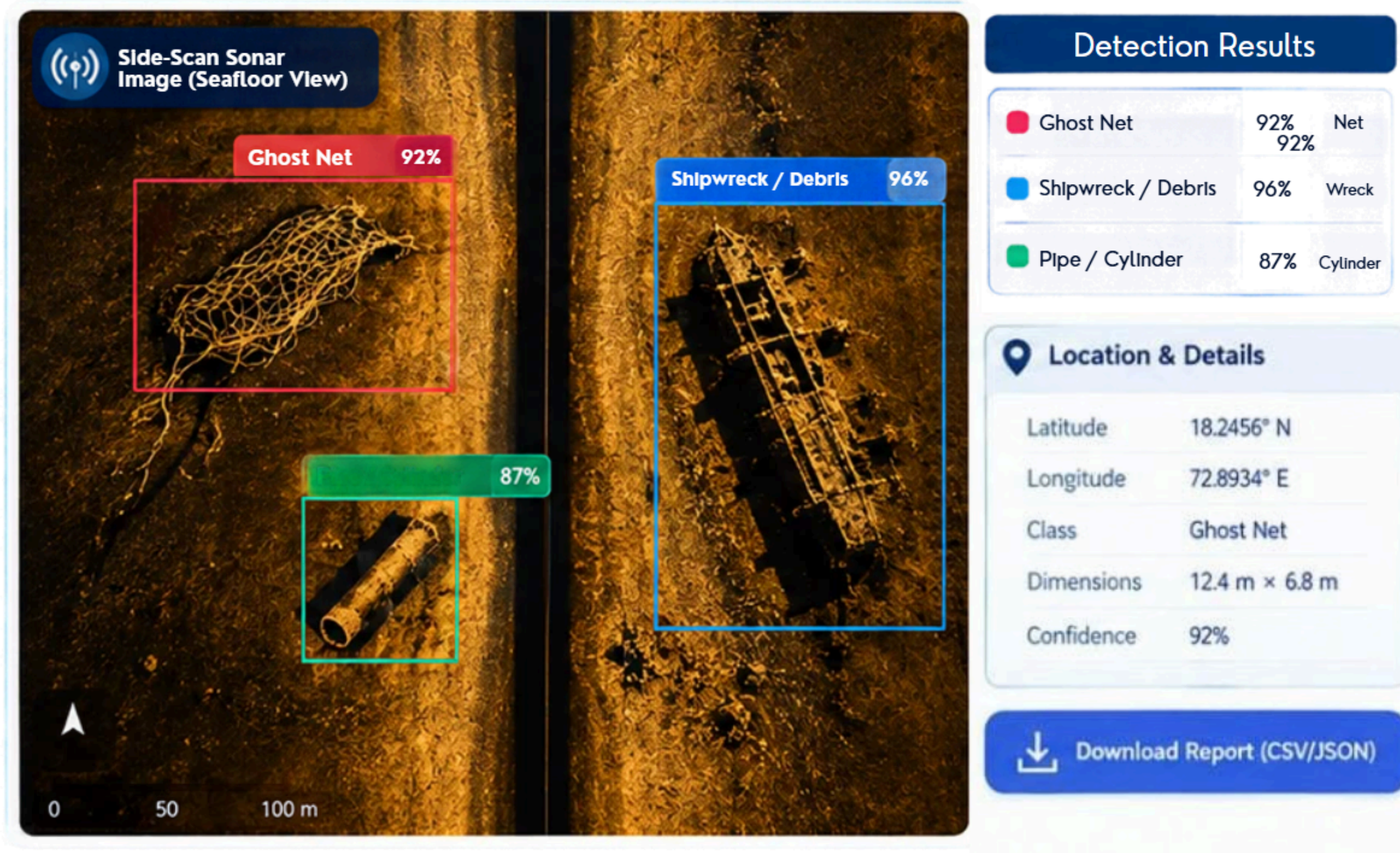


DATA & OPERATIONAL RISK

Data privacy / cyber security
Mitigation: Secure data handling & audit logs



MARINE DEBRIS DETECTION IN ACTION



IMPACTS



Marine Conservation

- Faster identification of ghost nets and debris
- Supports coral reef and marine-life protection
- Reduces hazardous debris exposure



Operational Efficiency

- Automates inspection of long sonar logs
- Reduces manual review time
- Supports AI/ship-based surveys



BENEFITS



Marine Life & Ecosystems

- Helps locate ghost nets and debris that threaten marine organisms and habitats.



Ocean Cleanup Teams

- Provides localized anomaly coordinates, dimensions, class and confidence for targeted recovery.



Research & Monitoring

- Enables structured CSV/JSON anomaly reports and consistent sonar-based monitoring.



Edge & Field Deployment

- Designed for efficient onboard/edge processing with reduced dependency on heavy cloud computing.

References

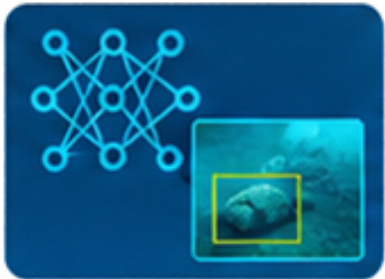
1 Side-Scan Sonar for Seafloor Mapping



- Side-scan sonar produces acoustic images of the seafloor.
- Used for identifying objects, debris, wrecks and seabed structures.
- Acoustic shadows help distinguish raised objects from the seabed.

Reference: NOAA Ocean Exploration – Sonar Technology

2 Underwater Object Detection using Deep Learning



- Deep learning can identify objects from complex underwater imagery.
- YOLO enables fast object detection suitable for near-real-time applications.
- U-Net can be used for pixel-level segmentation of debris and anomalies.



3 Marine Debris Detection & Classification



- AI-based vision systems can classify and localize underwater debris.
- Training data can include nets, pipes, cylinders, wrecks and other artificial objects.
- Data augmentation and noise filtering improve robustness against difficult underwater conditions.

Reference: Marine debris detection and underwater computer vision research

Comparison with Existing Systems

Automated Debris Detection			
Side-Scan Sonar Support	✗	✓	✓
Noise & Speckle Filtering	✓	✓	✓
Natural/Artificial Classification	✗	✓	✓
Confidence Score	✗	✓	✓
Bounding Box / Segmentation	✗	✓	✓
Geotagged Detection	✗	✗	✓
Anomaly Report Generation	✗	✗	✓
Real-Time Dashboard	✗	✓	✓
Edge/Onboard Deployment	✗	✓	✓
End-to-End Pipeline	✗	✗	✓

Prototype

Project Prototype:
<https://tarang-mdd.vercel.app>

GitHub Repository:
<https://github.com/iammyogesh/TARANG.git>

Existing AI-Based Marine/Underwater Systems



SeaClear

- All-assisted underwater litter detection and collection.
- Uses computer vision for identifying marine litter.



TrashCan Dataset

- Underwater trash image dataset for object detection and segmentation.
- Useful for training and evaluating marine debris detection models.



SONAR-based Object Detection Research

- Deep learning approaches have been explored for detecting objects in acoustic imagery.